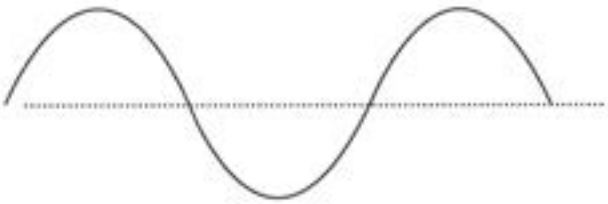
Vision I

Visible light is part of the electromagnetic spectrum and is made up of waves of different wavelength. Electromagnetic waves carry energy and travel in straight lines at 300 000 kilometres per second (km s^{-1}) in a vacuum. (A vacuum is an enclosed space with a total absence of any matter.) As the name implies, electromagnetic waves are electric and magnetic ripples caused by the vibration or movement of particles. In order of increasing energy, the electromagnetic spectrum includes: radio waves, microwaves, infra-red, visible light, ultra-violet, X-rays, gamma rays and cosmic rays. Their wavelengths start at 10^{8} metres, to smaller than 10^{-11} metres. Wavelength is the distance from peak-to-peak or trough-to-trough in a wave.

1 Copy the diagram of a wave, and indicate its wavelength on the diagram.



Visible light has wavelengths from around 400 nanometres (nm = 10^{-9} m) at the violet end of the spectrum to around 700 nm at the red end of the spectrum. Enormous numbers of animals have evolved specialised structures that are sensitive to visible light. The light receptors in many invertebrates don't form images in the same way that the human eye does, but just respond to the presence and intensity of light.

In vertebrates, eyes are the organs of sight, sending nervous impulses to the brain when stimulated by light rays. The brain interprets these impulses to produce images. Each eye consists of a hollow sphere (eyeball). It is set into a socket in the skull and is protected by eyelids and eyelashes. Muscles join the eyeball to the eye socket. They contract to enable the eye to swivel around. Glands at the top of each eye release a watery fluid that keeps the surface of the eyes moist and clean.

2 What do we call this 'watery fluid' that bathes the eyes?

Animals with complex eyes that can form images often see the world in a different fashion to humans. Their visual *acuity* (how sharply they can see things; keenness of perception) will also be different. The standard definition of normal human visual acuity is 20/20 vision. (This originally came from testing vision at a distance of 20 feet, or just over 6 metres.)

Humans, like most animals, have two eyes. Two eyes are enough to give us a good chance of survival in the wild. How the eyes are placed is also important. Animals such as rabbits, horses and birds have eyes on the sides of their head. This enables them to see in an almost complete circle and gives them a good chance to see predators sneaking up on them. However, side eyes are no good for judging distances. If both eyes are at the front of the head, it is called *binocular vision*. Each eye sees a slightly different version of the same object and so the brain can make fairly accurate estimate about how far away the object is, using the angle of the two eyes and by combining the images from each eye.

3 Give a definition of *predator* and give 3 examples.

Identify these parts of the human body.

4 A pair of spheres each measuring about 2.5 cm and set into sockets in the skull.

5 Large organ that coordinates stimuli from sense organs and acts on these stimuli.

6 Organ that secretes chemicals for use in the body.

7 Outline the differences in the vision of an animal with eyes at the side of its head compared with an animal with eyes at the front.

Research

8 Give two examples of how human vision differs from a named animal and give a possible reason for this difference.

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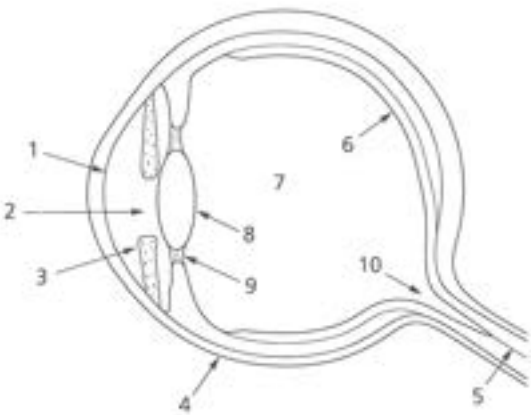
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The eye

The front part of your eye – the clear outer part – is the *cornea*. It is very sensitive and must be kept clean. Whenever your eyelids close, a watery fluid is released by the *lachrymal glands*. Tears contain salts and an enzyme that has an anti-bacterial action. This fluid helps to remove dust and other small particles that could irritate or damage the eye. Your eyebrows and eyelashes also play an important part in protecting your eyes from being damaged by water and small particles. If you get something in your eye, you should not rub it. This moves the particles further into the eye, making it more difficult to remove them. There are many times when you need extra protection for your eyes. Safety glasses should be worn when carrying out some science experiments or when using power tools.

Light enters the eye through the *cornea* and passes through the 'hole' in the *iris*, called the *pupil*. The size of the pupil is controlled by the iris to vary the amount of light passing into the eye. Light then passes through the *lens*, which focuses the light rays to a sharp image on the retina, or back of the eye. The *retina* is packed with light-sensitive cells (*rods and cones*). These cells send electrical impulses to the brain through a bundle of nerve fibres called the *optic nerve*. There are no rods and cones present in the area where the optic nerve leaves the eye. This area is called the *blind spot*. Receiving a sharp image on the retina is not all there is to seeing objects; the brain has to interpret the messages it receives from the retina cells.

9 Study the diagram of the eye. Identify the parts of the eye numbered 1 to 10, from the list:



blind spot, cornea, iris, lens, muscles, optic nerve, pupil, retina, sclera, vitreous humour

Many devices have been invented to extend our senses, in other words, to allow us to observe things that we would otherwise not be able to observe. Lenses are used in microscopes, telescopes and binoculars to extend our sense of sight when objects are very small or far away. Lenses are also used to correct faulty vision, such as short-sightedness (*myopia*), long-sightedness (*hyperopia*) and some cases of *cataracts*. A cataract is when the lens of the eye or its capsule becomes opaque. Patients with cataracts have diminished vision, but can usually distinguish light from darkness. The most common form of cataracts (senile) are associated with aging, causing the whole lens to become opaque. The most successful method of relieving senile cataracts is to remove the affected lens and replace it with an artificial lens.

10 What is meant by *opaque*?

Identify the following parts of the eye from their descriptions. (Questions 11 – 15)

11 Diaphragm that makes up the coloured part of the eye.

12 Sensory membrane that lines the inside of the eye.

13 Clear jelly that fills the inner part of the eyeball.

14 Nerve that carries electric pulses from retina to brain.

15 Dense white outer coat enclosing the eyeball excluding the cornea.

16 Name three cases when lenses can be used to aid the human eye.

17 Identify the glands that release tears.

18 Identify a use for tears in the eyes.

19 Identify two circumstances when eye protection should be worn.

Research

20 Find out what effect the 'blind spot' has on our vision.

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